Do Now · retrieval
Operating-system functions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does LMC LDA do?

Q2

What does the CPU control unit do?

Q3

Define a process.

Q4

Binary search reduces a sorted 32-element candidate set to one. How many successive halvings is this?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — The OS mediates application requests and access to hardware.

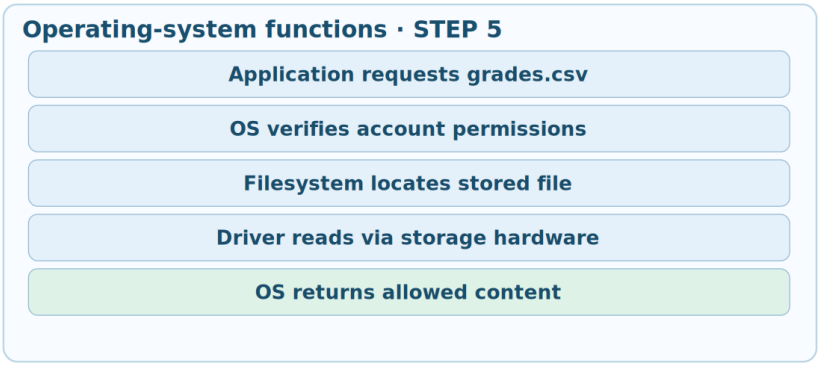
Core idea 2

Trace — Resource management schedules processes, allocates memory, controls files and drivers.

Core idea 3

Explain — Permissions and isolation help keep processes and users from accessing unauthorised data.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

process — Running/scheduled program instance

filesystem — Organised file storage

driver — Device interface software

spooling — Queued peripheral work

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A pupil app requests a protected file. Explain the OS steps.

Follow the worked method

1. Predict first — The OS checks authorisation before returning content; a missing permission should prevent the read.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The print job enters a queue and is later sent via a compatible printer driver when the device is available.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State four operating-system responsibilities. [4]

Question 2

Describe a protected file read from application request to result. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain how an OS manages a browser, game and backup concurrently, including protection. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Applications must directly control all hardware to read files.

Use key vocabulary from memory

1. Explain process in your own words.

2. Explain filesystem in your own words.

3. Explain driver in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.